

Mathematics Lecture Notes

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Chapter 1

Linear Algebra

This chapter introduces the basic objects and methods of linear algebra.

1.1 Vectors

This section will introduce vectors and their basic operations.

1.2 Matrices

This section will introduce matrices and basic matrix operations.

1.3 Systems of Linear Equations

This section will introduce systems of linear equations and their connection with matrices.

Chapter 2

Analytic Geometry

This chapter introduces geometric objects described using coordinates and algebraic equations.

2.1 Coordinates

This section will introduce coordinate systems and the representation of points.

2.2 Lines

This section will introduce equations of lines and their geometric interpretation.

2.3 Planes

This section will introduce equations of planes in three-dimensional space.

Chapter 3

Calculus

This chapter introduces functions, limits, continuity, and derivatives.

3.1 Functions

This section will introduce functions as mathematical objects describing dependence between quantities.

3.2 Limits

This section will introduce the concept of a limit.

3.3 Continuity

This section will introduce continuity and its basic interpretation.

3.4 Derivatives

This section will introduce derivatives as rates of change.